

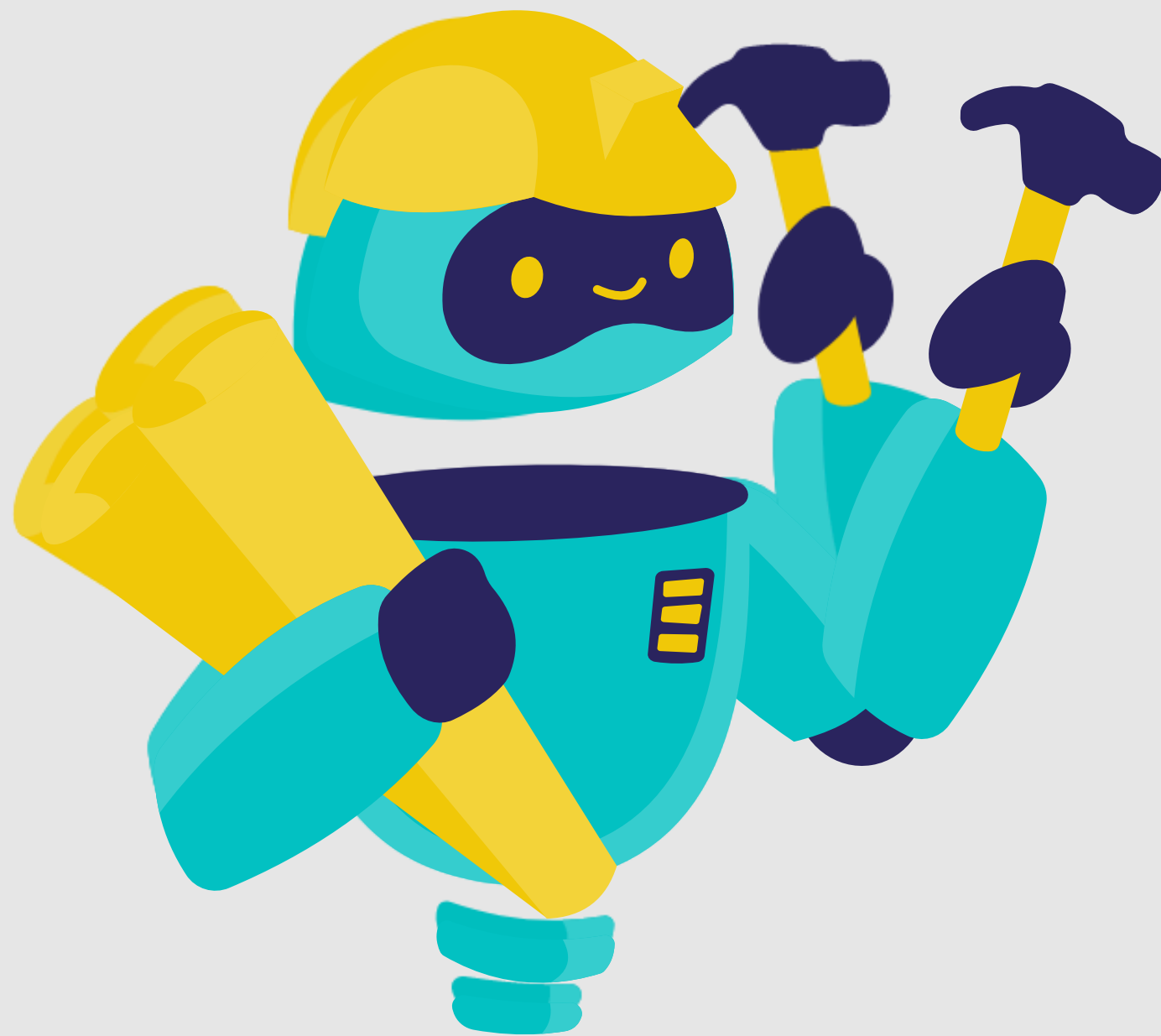


RENAD NASER TKROUNE

BUSINESS CASE: AUTOMATED CUSTOMER REVIEWS



INTRODUCTION



IN MANY COMPANIES, EVEN THOUGH CUSTOMER EXPERIENCE IS REGULARLY MEASURED, DATA TRANSFORMATION BECOMES CHALLENGING FOR MANY REASONS.

FOR INSTANCE, IT IS EXPENSIVE, RESOURCE-INTENSIVE, AND REQUIRES EXPERTISE. BUT THANKS TO MODERN TECHNOLOGIES LIKE MARTECH TOOLS AND THE USE OF ARTIFICIAL INTELLIGENCE (AI) IN CUSTOMER FEEDBACK ANALYSIS.

TYPES OF ARTIFICIAL INTELLIGENCE

- **SENTIMENT CLASSIFICATION (BERT-BASED)**

FINE-TUNED ON 15K REVIEWS TO CLASSIFY SENTIMENTS (E.G., POSITIVE, NEGATIVE)

- **REVIEW SUMMARIZATION (GPT-4)**

USES A TRANSFORMER TO GENERATE ARTICLE-STYLE SUMMARIES FOR EACH PRODUCT CATEGORY, HIGHLIGHTING TOP PRODUCTS, KEY FEATURES, AND MAJOR COMPLAINTS.

- **CLUSTERING MODEL (K-MEANS)**

GROUPS SIMILAR REVIEWS INTO CLUSTERS TO UNCOVER HIDDEN PATTERNS AND TOPICS—GREAT FOR EXPLORING THEMES WITHOUT LABELED DATA.



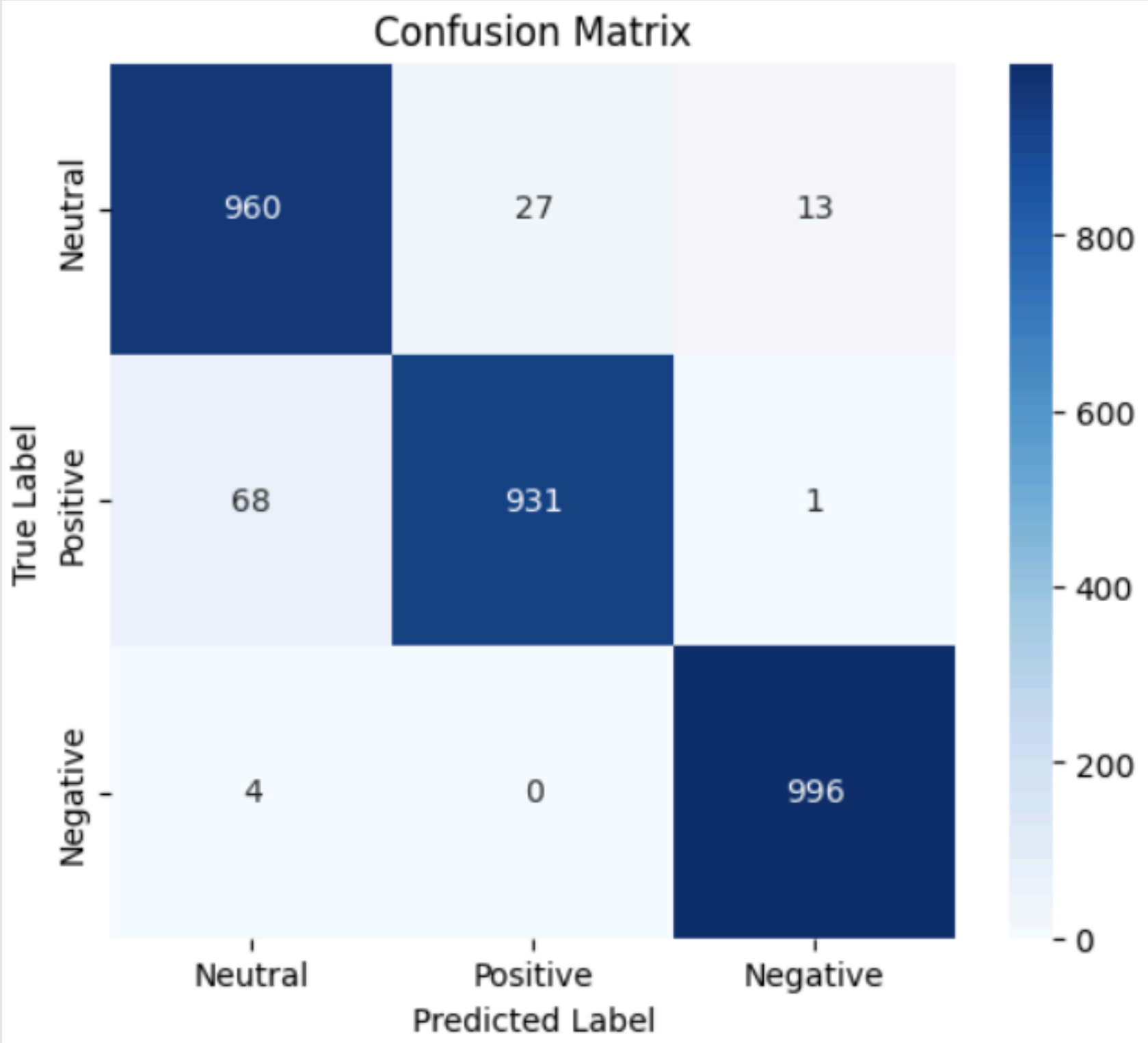
- EXPLORED DATASET STRUCTURE AND CONTENT TO UNDERSTAND FORMATS, DISTRIBUTIONS, AND POTENTIAL ISSUES
- REMOVED UNNECESSARY COLUMNS THAT DIDN'T CONTRIBUTE TO MODEL LEARNING
- COMBINED RELEVANT COLUMNS (TITLE + REVIEW TEXT) TO PROVIDE RICHER CONTEXT AND IMPROVE MODEL PERFORMANCE
- HANDLED NULL VALUES BY REMOVING ROWS OR REPLACING WITH MEANINGFUL DEFAULTS
- STANDARDIZED NUMERICAL VALUES TO ENSURE CONSISTENCY ACROSS ALL SAMPLES (RATINGS SCALED FROM 1-5)
- CLEANED TEXT DATA BY:
 - LOWERCASING ALL TEXT
 - REMOVING SPECIAL CHARACTERS, EXTRA WHITESPACE, AND STOP WORDS
 - HANDLING CONTRACTIONS (E.G., "DON'T" → "DO NOT")
- TOKENIZED AND ENCODED TEXT USING APPROPRIATE TOKENIZERS
- CHECKED CLASS BALANCE AND PERFORMED RESAMPLING IF NECESSAR



DATA PROCESSING & CLEANING

SENTIMENT CLASSIFICATION:

MODEL 1: DISTILBERT-BASE-UNCASED



- **MODEL:** DISTILBERT-BASE-UNCASED
- **DATASET:** 50,000 LABELED PRODUCT REVIEWS
- **CLASSES:** POSITIVE, NEGATIVE, NEUTRAL (MULTI-CLASS CLASSIFICATION)
- **ACCURACY:** 96% ON VALIDATION SET
- **PREPROCESSING:** TOKENIZATION WITH BERT TOKENIZER, LOWERCASING, SPECIAL TOKEN HANDLING
- **FINE-TUNING:** USED PRE-TRAINED BERT BASE MODEL WITH CLASSIFICATION HEAD
- **OPTIMIZER:** LEARNING RATE ADJUSTMENT
- **USE CASE:** AUTOMATES SENTIMENT TAGGING FOR REVIEWS, USEFUL IN DASHBOARDS, ALERTS, AND ANALYTICS

SENTIMENT CLASSIFICATION:

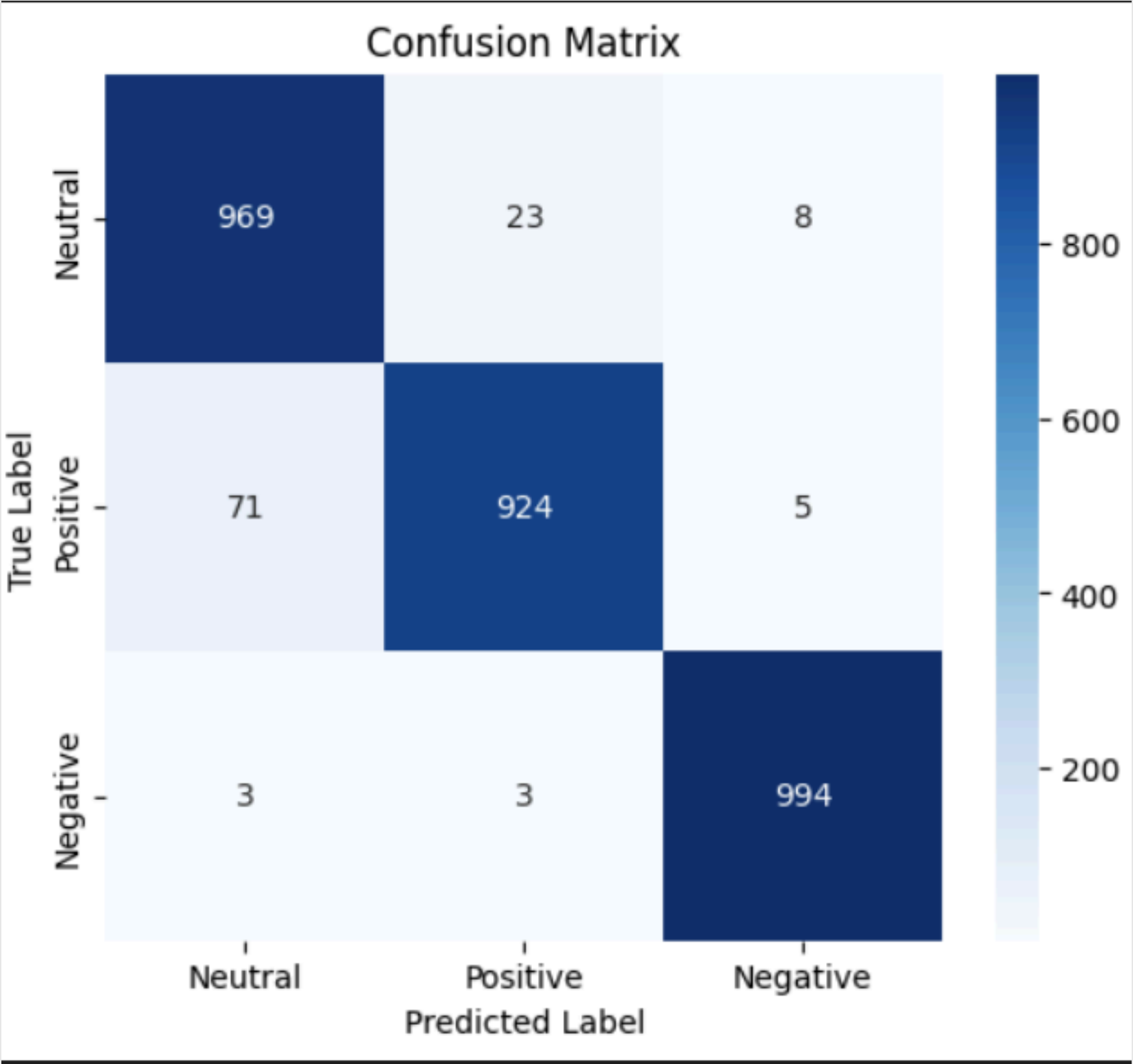
MODEL 1: DISTILBERT-BASE-UNCASED

Epoch	Training Loss	Validation Loss	Accuracy	F1	Precision	Recall
1	0.589500	0.296927	0.902000	0.901968	0.901958	0.902000
2	0.201000	0.202221	0.940333	0.940361	0.941638	0.940333
3	0.112300	0.169736	0.961333	0.961276	0.961283	0.961333
4	0.058900	0.172140	0.962333	0.962299	0.962729	0.962333

	precision	recall	f1-score	support
Neutral	0.93	0.96	0.94	1000
Positive	0.97	0.93	0.95	1000
Negative	0.99	1.00	0.99	1000
accuracy			0.96	3000
macro avg	0.96	0.96	0.96	3000
weighted avg	0.96	0.96	0.96	3000

SENTIMENT CLASSIFICATION:

MODEL 2: BERT-BASE-UNCASED



- **MODEL:** BERT-BASE-UNCASED
- **DATASET:** 50,000 LABELED PRODUCT REVIEWS
- **CLASSES:** POSITIVE, NEGATIVE, NEUTRAL (MULTI-CLASS CLASSIFICATION)
- **ACCURACY:** 96% ON VALIDATION SET
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SENTIMENT CLASSIFICATION:

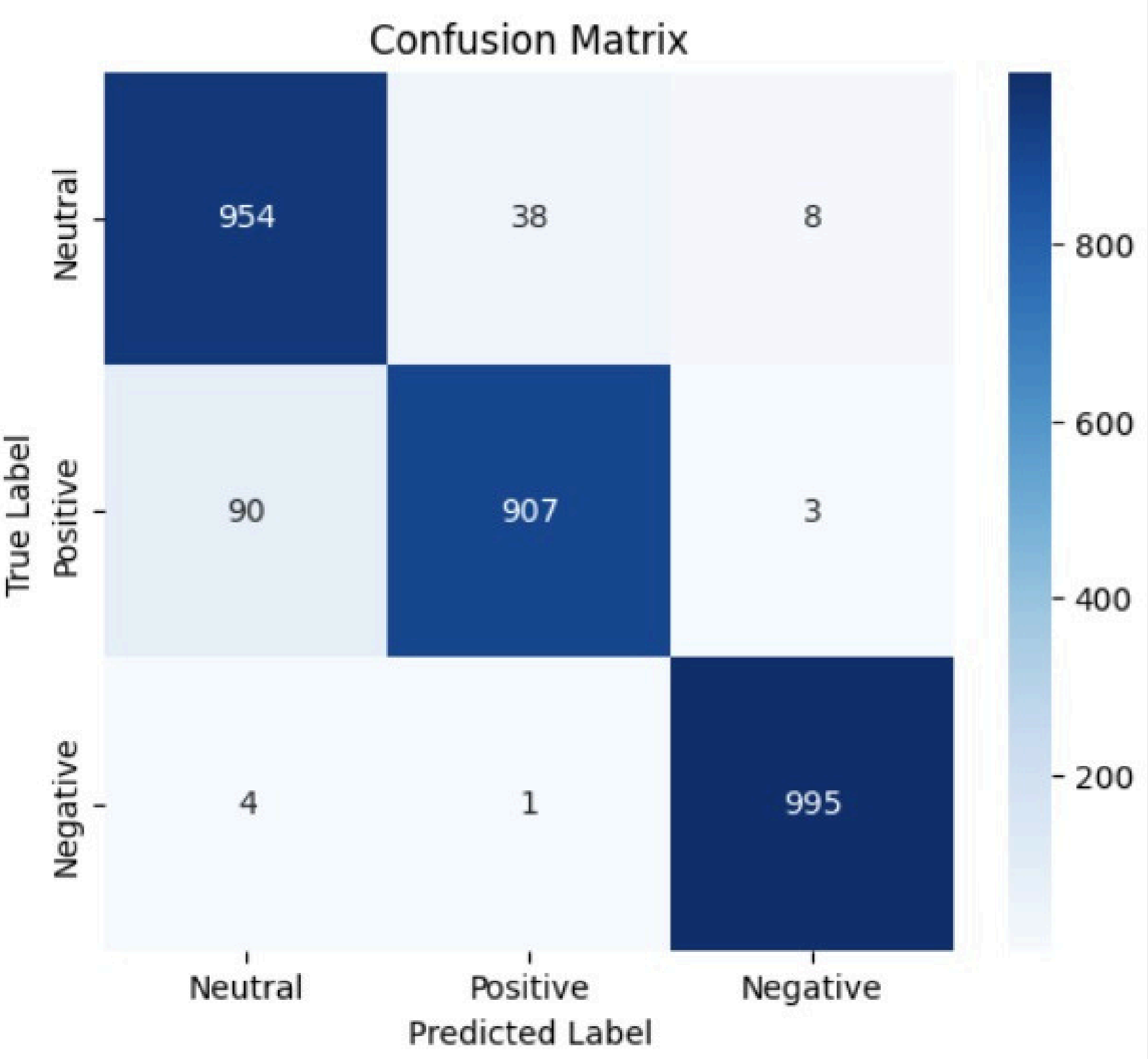
MODEL 2: BERT-BASE-UNCASED

	precision	recall	f1-score	support
Neutral	0.93	0.97	0.95	1000
Positive	0.97	0.92	0.95	1000
Negative	0.99	0.99	0.99	1000
accuracy			0.96	3000
macro avg	0.96	0.96	0.96	3000
weighted avg	0.96	0.96	0.96	3000

Epoch	Training Loss	Validation Loss	Accuracy	F1	Precision	Recall
1	0.592000	0.308134	0.895333	0.895676	0.896385	0.895333
2	0.171300	0.215572	0.949000	0.949080	0.950297	0.949000
3	0.087400	0.213241	0.959000	0.958929	0.959660	0.959000
4	0.034900	0.215861	0.962333	0.962277	0.962924	0.962333

SENTIMENT CLASSIFICATION:

MODEL 3: ROBERTA-BASE



- **MODEL:** ROBERTA-BASE
- **DATASET:** 50,000 LABELED PRODUCT REVIEWS
- **CLASSES:** POSITIVE, NEGATIVE, NEUTRAL (MULTI-CLASS CLASSIFICATION)
- **ACCURACY:** 95% ON VALIDATION SET
- **PREPROCESSING:** TOKENIZATION WITH BERT TOKENIZER, LOWERCASING, SPECIAL TOKEN HANDLING
- **FINE-TUNING:** USED PRE-TRAINED BERT BASE MODEL WITH CLASSIFICATION HEAD
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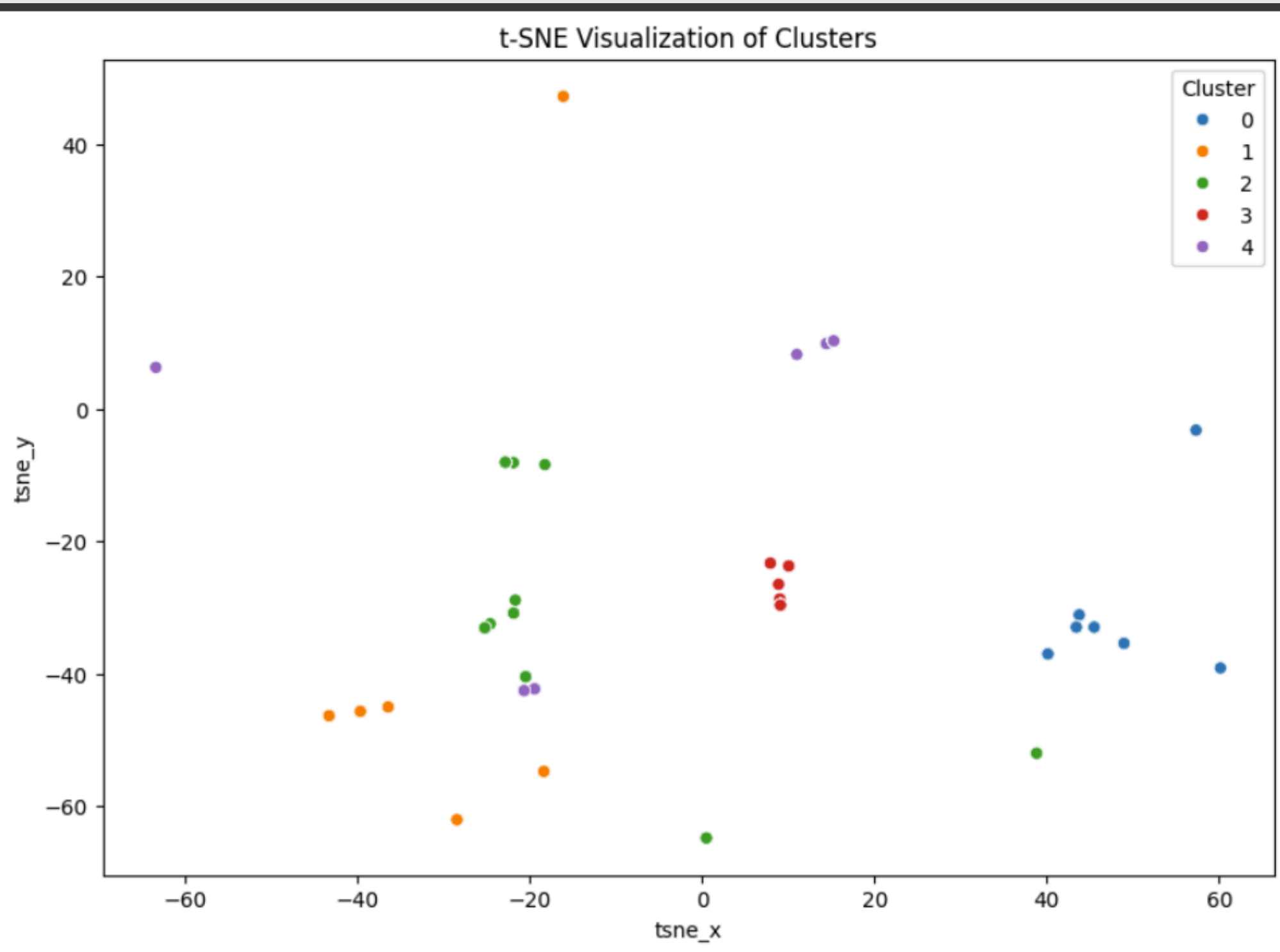
SENTIMENT CLASSIFICATION:

MODEL 3: ROBERTA-BASE

Epoch	Training Loss	Validation Loss	Accuracy	F1	Precision	Recall
1	0.585900	0.325945	0.886333	0.887145	0.888814	0.886333
2	0.209900	0.256869	0.929333	0.929342	0.929632	0.929333
3	0.134200	0.282668	0.949333	0.949297	0.949309	0.949333
4	0.065300	0.290449	0.952000	0.951944	0.952715	0.952000

	precision	recall	f1-score	support
Neutral	0.91	0.95	0.93	1000
Positive	0.96	0.91	0.93	1000
Negative	0.99	0.99	0.99	1000
accuracy			0.95	3000
macro avg	0.95	0.95	0.95	3000
weighted avg	0.95	0.95	0.95	3000

CLUSTERING WITH K-MEANS

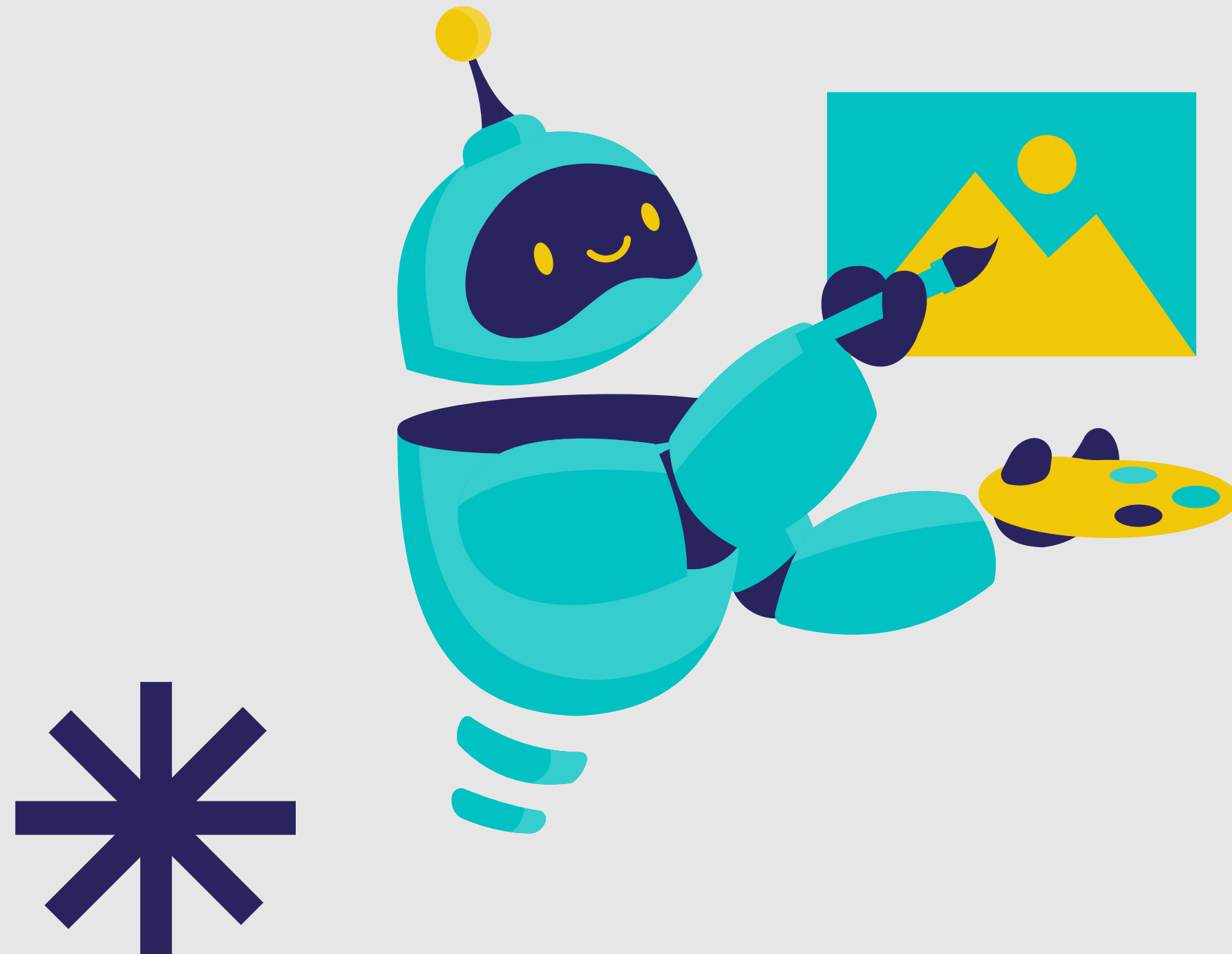


- **MODEL:** K-MEANS CLUSTERING ALGORITHM
- **INPUT:** VECTORIZED REVIEW TEXTS (E.G., USING TF-IDF OR BERT EMBEDDINGS)
- **OUTPUT:** GROUPS OF SIMILAR REVIEWS (CLUSTERS)
- **GOAL:** DISCOVER UNDERLYING THEMES AND RECURRING TOPICS WITHOUT LABELED DATA
- **PREPROCESSING:** TEXT CLEANING, DIMENSIONALITY REDUCTION (E.G., PCA)
- **USE CASE:** IDENTIFIES HIDDEN INSIGHTS LIKE COMMON FEATURE REQUESTS OR PAIN POINTS
- **BENEFITS:** GREAT FOR EXPLORATORY ANALYSIS AND ORGANIZING LARGE-SCALE FEEDBACK

REVIEW SUMMARIZATION (GPT-4)

- **MODEL:** OPEN AI GPT-4
- **INPUT:** LARGE BATCHES OF PRODUCT REVIEWS GROUPED BY CATEGORY
- **OUTPUT:** ARTICLE-STYLE SUMMARIES INCLUDING:
 - TOP 3 PRODUCTS
 - MOST COMMON RATINGS
 - WORST-RATED PRODUCT
- **PURPOSE:** CONVERTS RAW REVIEW DATA INTO CONCISE, INSIGHTFUL SUMMARIES
- **BENEFITS:** SAVES TIME FOR USERS, HIGHLIGHTS KEY TRENDS, SUPPORTS PRODUCT COMPARISONS
- **TECHNIQUES:** PROMPT ENGINEERING, SUMMARIZATION TUNING, CHUNKING LONG INPUTS

BENIFETS



- CLUSTERING REVEALS UNDERLYING THEMES AND USER CONCERNS THAT AREN'T OBVIOUS AT FIRST GLANCE.
- SUMMARIZED FEEDBACK HELPS CUSTOMERS CHOOSE BETTER PRODUCTS AND BUSINESSES ADDRESS KEY ISSUES FASTER.
- ANALYZE THOUSANDS OF REVIEWS IN SECONDS—FAR BEYOND WHAT MANUAL ANALYSIS CAN HANDLE.

LOW MODEL ACCURACY

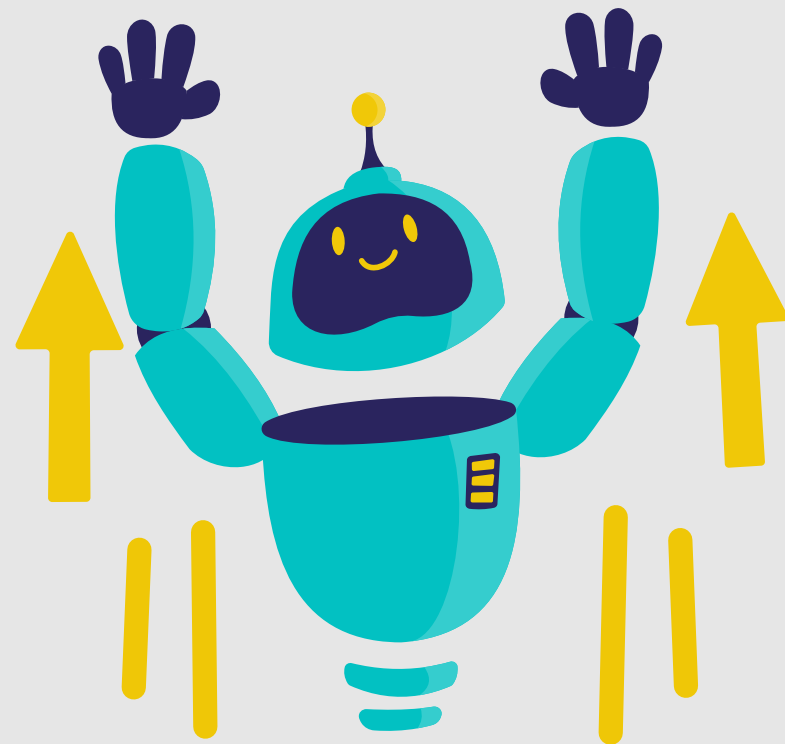
Challenge: Initial models achieved low accuracy due to noisy inputs and imbalanced classes

SOLUTION:

Cleaned and standardized review text

Balanced dataset with resampling techniques

Tuned hyperparameters (learning rate, batch size)



CHALLENGES

DEPLOYMENT AWS SAGEMAKER

Challenge: Packaging the model and setting up inference file

SOLUTION:

Try another deployment server but unfortunately a technical problem has appeared

DIFFICULTY INTERPRETING CLUSTERS

Challenge: K-Means clusters weren't clearly labeled or easy to understand

SOLUTION:

Analyzed top categories per cluster

Labeled clusters manually

IN-CODE TESTING

CLUSTERING WITH K-MEANS

```
def predict_category(text):
    emb = model.encode([text])
    pred = clf.predict(emb)
    cluster_names = {
        2: 'Tablets & eReaders',
        0: 'Holiday Shop',
        4: 'Fire TV',
        3: 'Frys',
        1: 'Computers & Tablets'
    }

    result = cluster_names.get(pred[0], "Unknown category") # Use get() to handle any out-of-bound values
    return result

# Try it:
#predict_category("Amazon Fire HD Tablet")
predict_category(["Considered by the Irish Times as one of the most brilliant flights of pure fancy\
ever recorded, this amazing story explores the beauty and untamed nature of a faraway world, where wild creatures crowd the f
s landscape and demented torturers dominate victims with their bizarre mental powers."])
```

'Tablets & eReaders'

IN-CODE TESTING

CLUSTERING WITH K-MEANS

```
for category, summary in category_summaries.items():
    prompt = build_prompt(category, summary)
    response = get_completion(prompt)
    print(response)
```

🔍 Top 3 Products:

- . Amazon 5W USB Official OEM Charger and Power Adapter for Fire Tablets and Kindle eReaders
 - Key Features: This is an official Amazon product, ensuring compatibility and safety for your devices. It provides 5W, 1A power output.
 - Strengths: Users appreciate its reliability and efficiency. It charges devices quickly and is compact, making it easy to carry around.
 - Common Complaints: Some users have reported that the charger stopped working after a few months of use.
- . Kindle Dx Leather Cover, Black (fits 9.7 Display, Latest and 2nd Generation Kindle Dxs)
 - Key Features: This cover is designed specifically for the Kindle DX with a 9.7-inch display. It's made of genuine leather and provides protection.
 - Strengths: Users love the high-quality feel of the leather and the snug fit for their Kindle DX. It also has a stylish design.
 - Common Complaints: Some users have reported that the cover can be a bit bulky and heavy.
- . Amazon Fire TV
 - Key Features: Amazon Fire TV provides access to thousands of channels, apps, and Alexa skills. It supports 4K Ultra HD, Dolby Vision, and HDR10.
 - Strengths: Users love the easy setup and user-friendly interface. It also has excellent picture quality and sound.
 - Common Complaints: Some users have reported issues with the remote control not working properly.

⚠️ One to Avoid:

Amazon Kindle Fire 5ft USB to Micro-USB Cable (works with most Micro-USB Tablets)

This product has received numerous complaints from users. Many have reported that the cable stopped working after a short period of use.



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THANK YOU